

ICT - Semester 7

NATURAL LANGUAGE PROCESSING (20IC403T)

Dr. Jigar Shah
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Syllabus:

UNIT-1: OVERVIEW OF NLP

Basics of Speech Processing; Place and Manner of Articulation; Word Boundary Detection; Argmax based computations; HMM and Speech Recognition, Language Structure and Analyzer - Overview of language, requirement of computational grammar. Words and their Analysis. Tokenization. Stemming. Morphological Analysis. POS tagging.

UNIT 2 CORPUS AND DATASETS

Corpus, need for corpus, corpus analysis, types of data attributes, Categorical or qualitative data attributes, Numeric or quantitative data attributes, different file formats for corpora, dataset preparation for NLP applications.

UNIT 3 SENTENCE STRUCTURE AND PROCESSING

NLP components, understanding, Morphological analysis, syntactic analysis, Semantic analysis, ambiguity handling, pragmatic analysis, Handling corpus-raw text, sentences, pre-processing.

UNIT 4 FEATURE EXTRACTION AND ADVANCED ALGORITHMS

Parsers and parsing, POS tagging and POS taggers, Name entity recognition, n-grams, bag of words, semantic tools and resources, probabilistic theory for NLP, TF-IDF, Vectorization, Encoders and decoders, normalization, Recall word embedding, Algorithms used by neural networks, Rule-Based System for NLP, Machine Learning for NLP Problems, Deep learning for NLP, case studies.

Some of the above topics would be covered through the associated laboratory course.

Teaching Scheme:

L	T	P	C	Hrs./Week	Theory			Practical		Total Marks	
					MS	ES	IA	LW	LE/Viva		
2	0	0	2	2	25	50	25	--	--	100	

Pre-requisite:

- Probability and Statistics
- Machine Learning
- Deep Learning

➤ Some Topics are covered here at required level

Reference Books (Theory):

- Jurafsky, Daniel, and James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics, Prentice Hall. 2026 Edition.
- Yoav Goldberg, Graeme Hirst. Neural Network Methods in Natural Language Processing, Morgan & Claypool Publishers.
- Kamath, Uday, et al. Large Language Models: A Deep Dive.

Reference Book (Hands-on Practical):

- Bird, Steven, Ewan Klein, and Edward Loper. Natural language processing with Python: analyzing text with the natural language toolkit. O'Reilly Media, Inc.
- Thanaki, Jalaj. Python natural language processing. Packt Publishing Ltd.
- Alammam, J., & Grootendorst, M. Hands-On Large Language Models, O'Reilly.

Online Video Course and Support Materials:

- NPTEL Course: Deep Learning for Natural Language Processing, IIT Kharagpur, Prof. Pawan Goyal
 - <https://nptel.ac.in/courses/108105572>
 - <https://sites.google.com/view/dl4nlp-nptel/home>

NATURAL LANGUAGE PROCESSING

ICT - Semester 7

UNIT 1

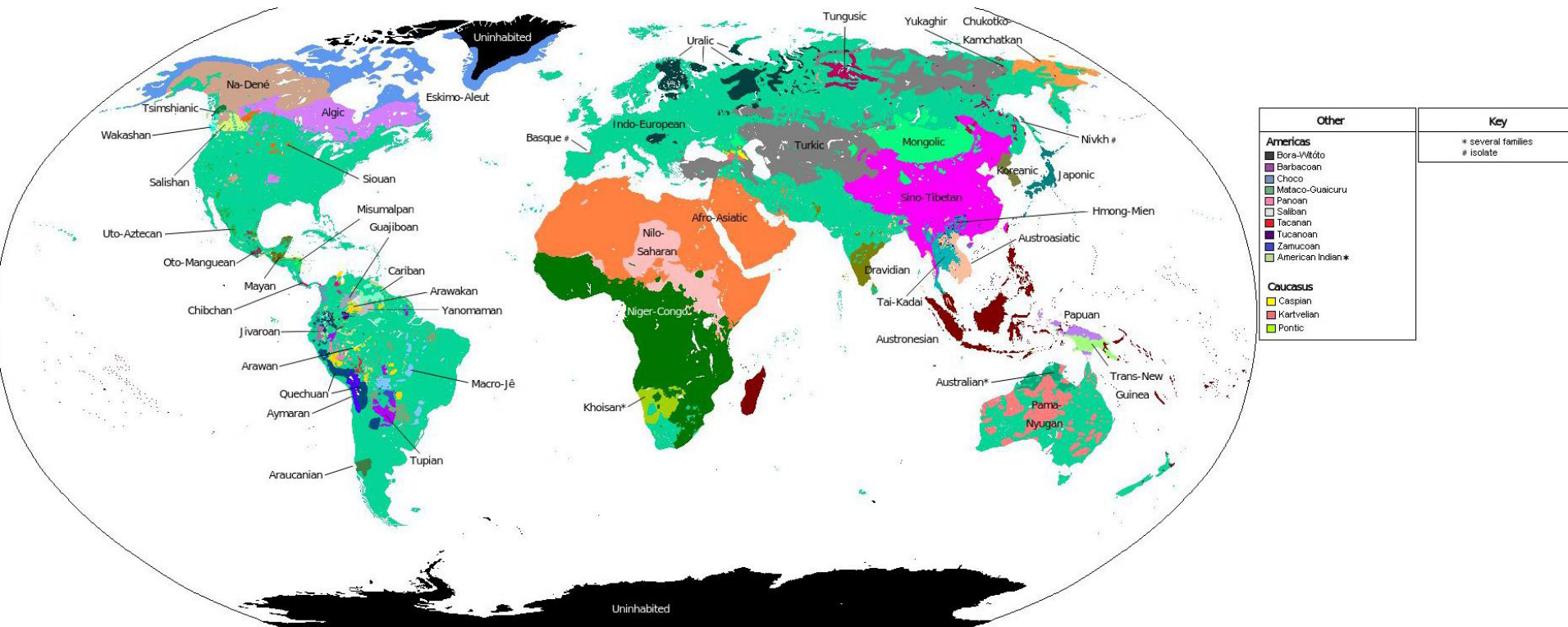
OVERVIEW OF NLP

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Introduction to NLP

What is Natural Language?

Natural Languages: Languages that evolved naturally through human use



What is NLP?

- NLP:
 - *Making computers understand what we write (or speak)*
 - *Making computers write (and speak)*
- Allows computers to interpret, manipulate, and comprehend human language.
- The field of NLP attempts to design, implement and test systems that process natural languages for practical applications like text translation, text summarization, speech to text conversion, chatbots etc.
- Conventional NLP refers to the rule-based and statistical approaches to processing human language.
- Now it is a subfield of Artificial Intelligence used to aid computers in understanding natural human language.
- Most NLP techniques rely on machine learning/deep learning/reinforcement learning to derive meaning from human languages
- Deep learning-based NLP refers to approaches that use neural networks—particularly deep architectures like recurrent, convolutional, and transformer models—to process and understand human language.

NLP Tasks

- Sentiment Analysis
- Machine Translation
- Text Classification
- Text Summarization
- Question Answering
- Text Entailment
- Named Entity Recognition (NER)
- Chatbots
- Language Modeling

NLP Tasks

Sentiment Analysis

Sentence

Sentiment

“I absolutely love this phone!”

Positive

“This restaurant is terrible.”

Negative

“The movie was okay, nothing special.”

Neutral

“I have never tasted a pizza like this”

.....



NLP Tasks

Text Classification

Text

Label

“Win a free iPhone now!
Click here!”

Task
Spam Detection

Spam

“Your order has been shipped.”

Task
Spam Detection

Not Spam

“The stock market closed higher
today after...”

Task
Topic Classification

Business/Finance

Cristiano Ronaldo scores a
hat-trick...”

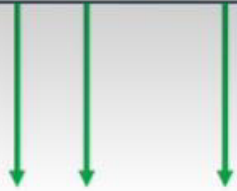
Task
Topic Classification

Sports

NLP Tasks

Machine Translation

I like Mangoes



Ich mag Mangos

Sequence of words in the input and output sentence are same

Translate to
German



I don't like Mangoes



Ich mag keine Mangos

Sequence of words in the input and output sentence are different

NLP Tasks

Text Summarization

Original Text

“NASA’s James Webb Space Telescope captured detailed images of a distant exoplanet’s atmosphere, revealing evidence of water vapor, clouds, and possible signs of life-supporting conditions. Scientists say this is a major breakthrough in the search for habitable worlds.



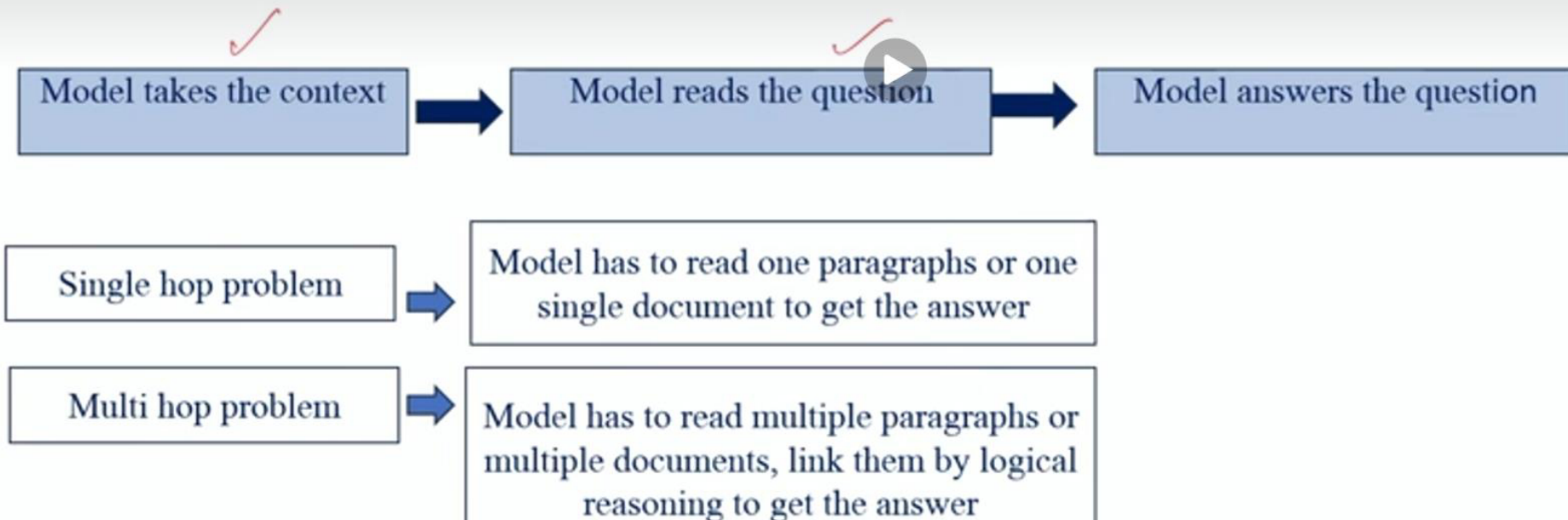
Summary

“NASA’s James Webb Telescope found water vapor and clouds on a distant exoplanet, a big step in the search for habitable worlds.”

NLP Tasks

Question Answering

Question	Answer
“Who was the first woman to win a Nobel Prize?”	Marie Curie



NLP Tasks

Text Entailment

Two sentences will be given- Model have to tell if these sentences contradict, entail or neutral

Example:

A black race car starts at the front of the crowd ✓

A black race car starts at the front of a lonely road ✓

Contradicts

A man is riding a bicycle in the park.

A person is cycling outdoors.

Entails

He is playing cricket with the kids ✓

The kids are enjoying ice cream ✓

Neutral

NLP Tasks

Chatbot

Customer Support Chatbot

User

Chatbot

- Hi, I ordered a phone last week. Can you check its delivery status?
- It's #54732.
- Great. Can I change the delivery address?

✓ Sure! Could you provide your order ID, please?



- ✓ Thanks! Let me check... Your phone was shipped yesterday and is expected to arrive on Friday.
- ✓ Yes, you can. Please click this link to update your address before 6 PM today.

NLP Tasks

Named Entity Recognition (NER)

NER detects **real-world objects** mentioned in text and labels them with their entity type.

Sourav Ganguly served as the captain of Indian Cricket team and he is from Kolkata

NER Output:

- Sourav Ganguly → **PERSON**
- Indian Cricket team → **ORGANIZATION**
- Kolkata → **LOCATION**



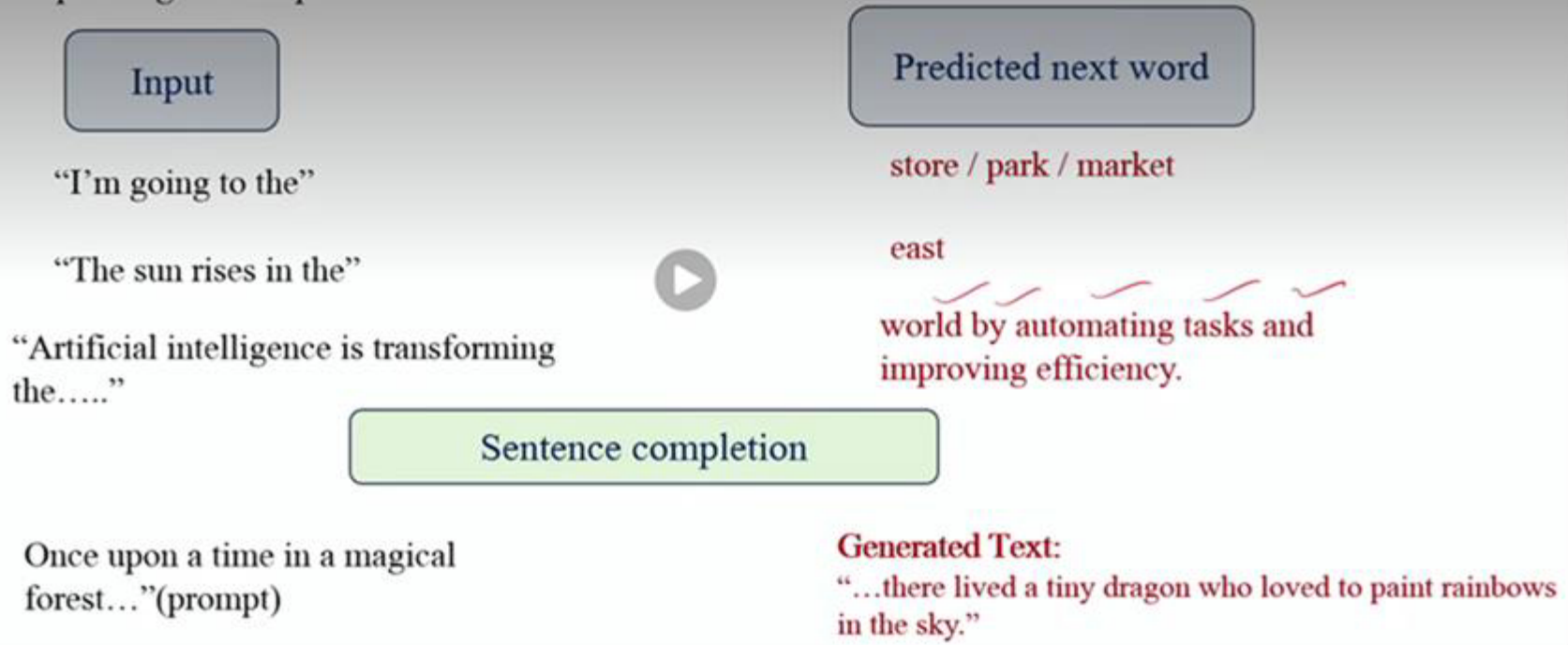
Other NER includes

- **TIME** – specific times
- **DATE** – date expressions
- **MONEY** – monetary values
- **PERCENT** – percentages
- **FACILITY** – buildings, airports
- **EVENT** – historical or social events

NLP Tasks

Language Modeling

Language modeling in NLP is the use of statistical and machine learning techniques to determine the probability of a sequence of words given the previous words.



- A Large Language Model (LLM) is an advanced artificial intelligence system designed to understand, process, and generate human-like text. It works by predicting the most logical next words in a sequence based on vast amounts of training data. Popular examples include OpenAI's GPT models and Google's Gemini.
- LLMs are giant statistical models built on deep learning and a specific neural network architecture known as the transformer. These networks have billions of "parameters" (internal variables) that act much like a human brain's synapses.

Challenges in NLP

Discrete Nature of
Text data

Sky and stars
Red and pink



Some relation is there among these pair of words
but words doesn't tell anything about the similarity

Evolving nature of
Natural Language

New words gets added to the dictionary (quaratene) time to time

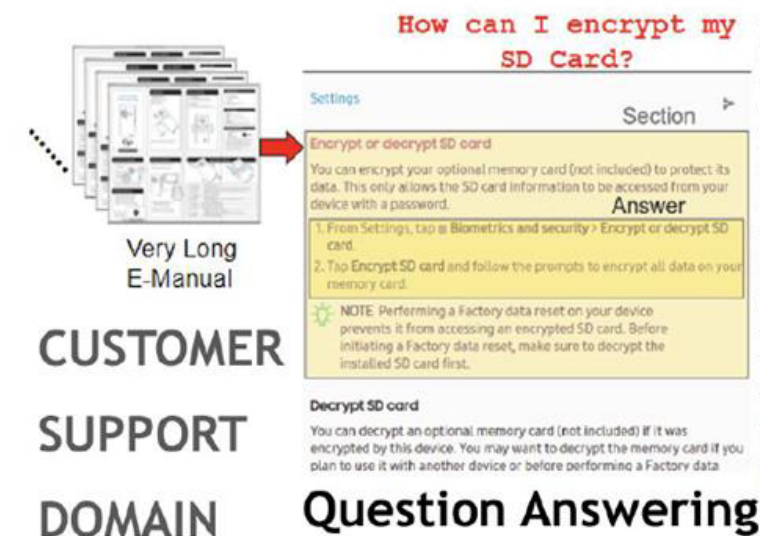
Ambiguity/Vagueness

Never tasted a pizza like this before

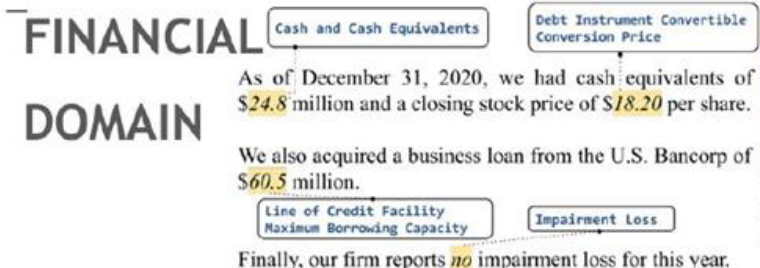
Capturing long range
dependency

Animal did not cross the road as it was (wide/tired)

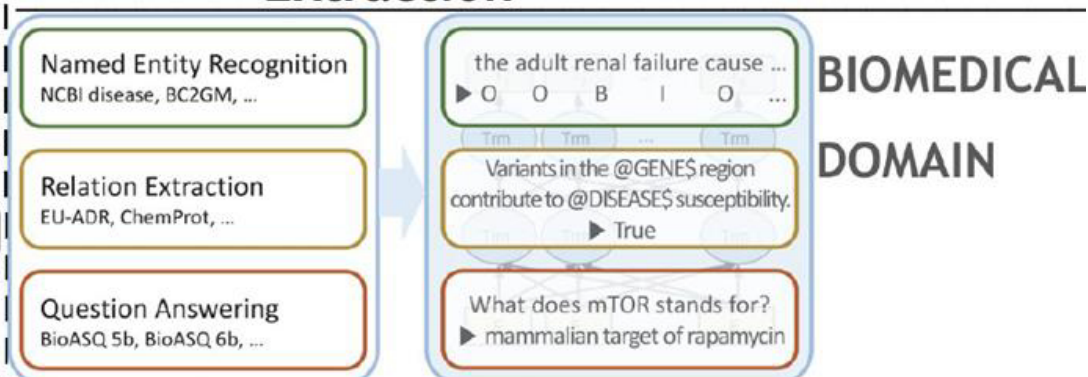
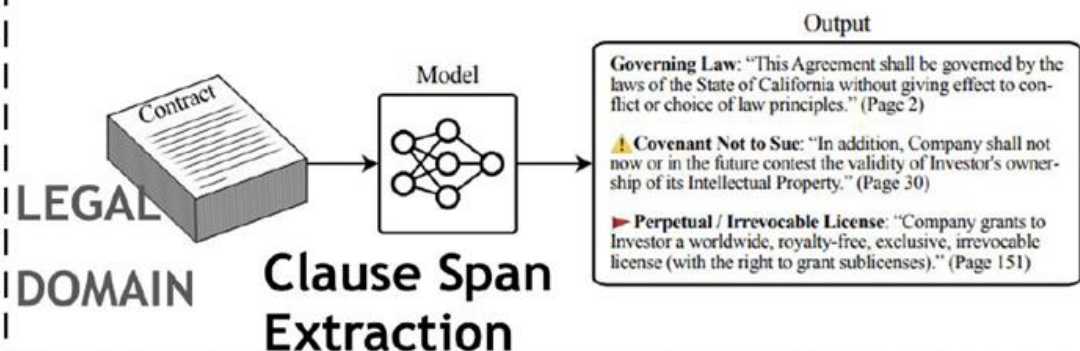
Domain Specific Applications



Question Answering



NER

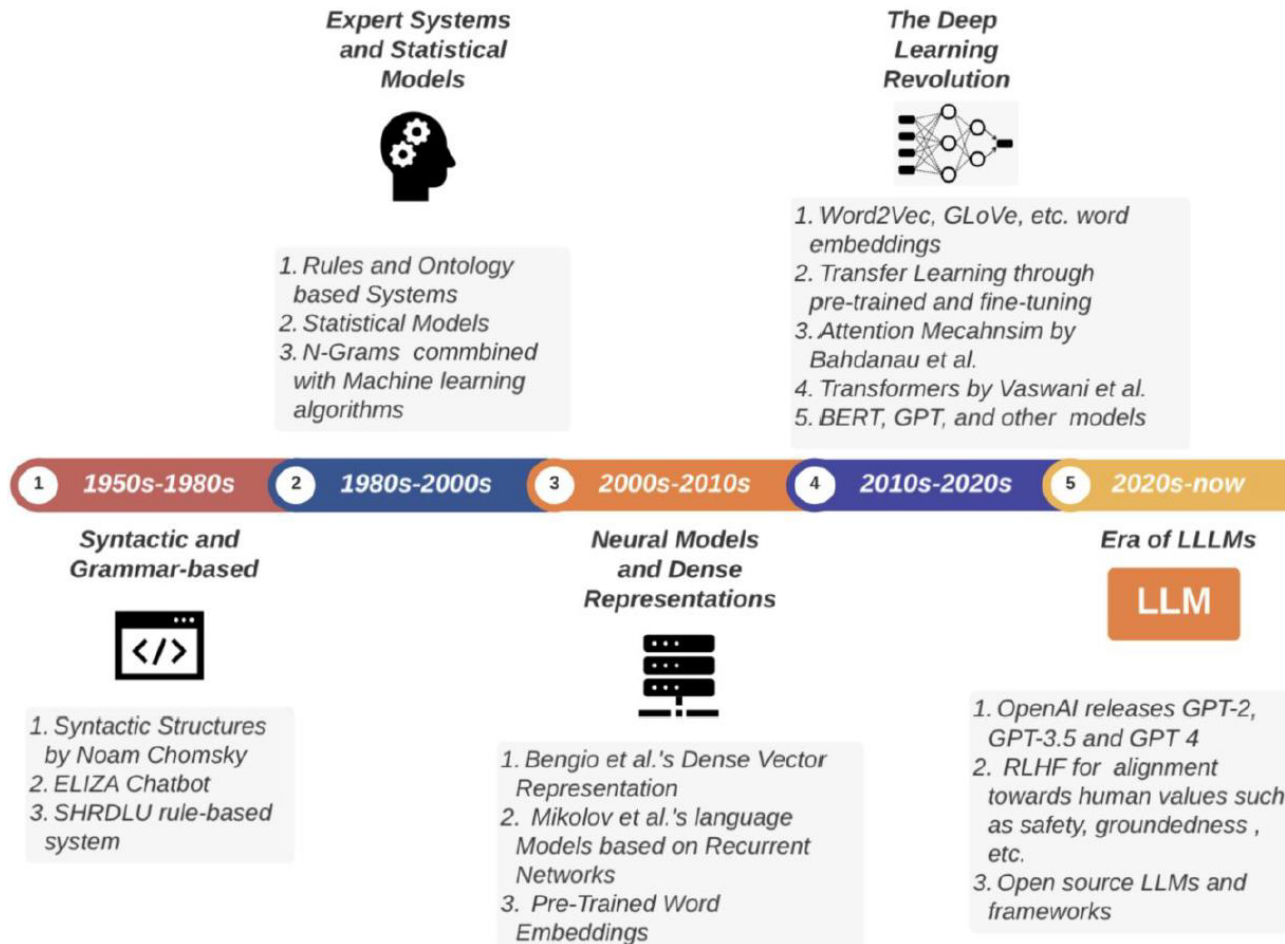


Several sentence and token level tasks

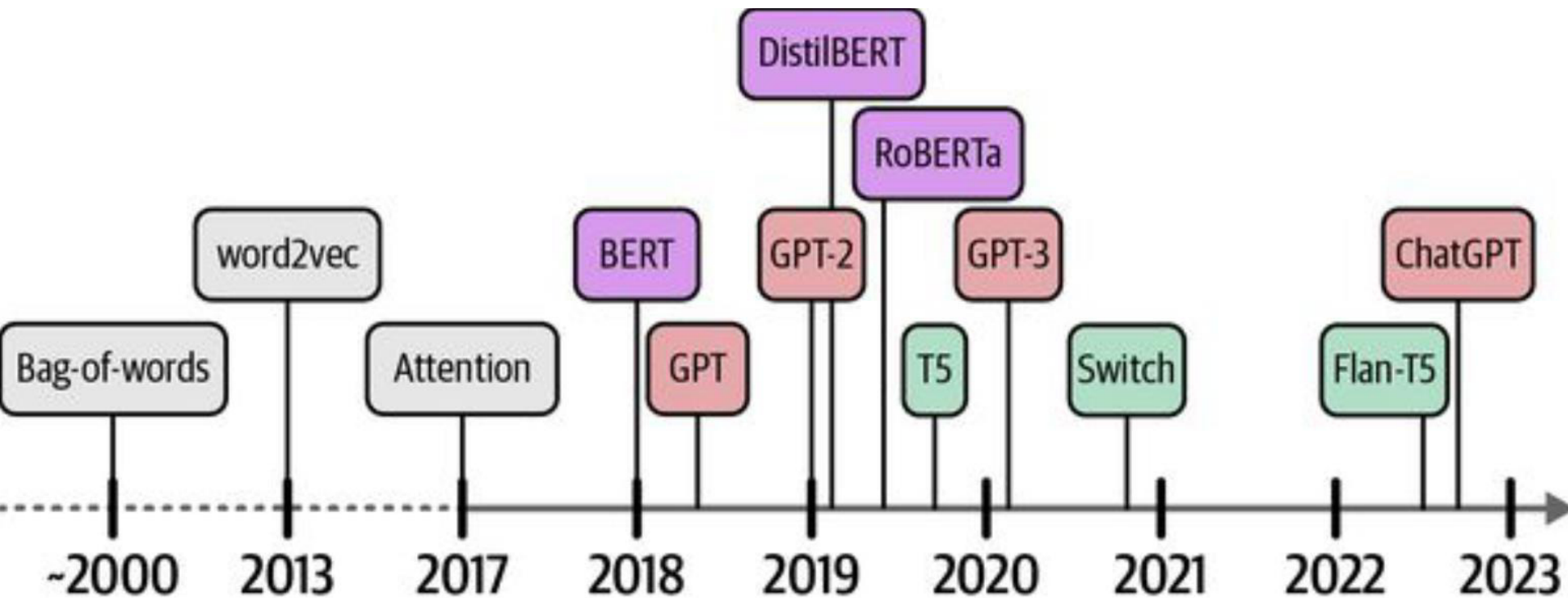
NLP Paradigms

- We generally try to map problems to various ML paradigms
 - Sentiment Analysis, news article groupings, etc. → Text Classification
 - Named entity recognition, code-mixing, etc. → Sequence Labelling
 - Machine Translation, summarization, chatbots, etc. → Text Generation

NLP Progression Timeline



A timeline of the recent developments



BERT stands for Bidirectional Encoder Representations from Transformers

GPT stands for Generative Pre-trained Transformer